

TPP-DLW Toolpath Generator

New User Guide for STL-to-TXT Conversion

This guide explains how to convert an STL file into the tab-separated segment text format used by the LabVIEW writing interface. It focuses on the fastest current workflow and the checks that prevent false toolpath lines.

Recommended path: use `heightmap_raster_export.m` for pixel-exact height-map STL/CSV files. Use the modular `tppdlw_process` pipeline for general STL/STEP workflows or the GUI.

1. Project Layout

Height-map raster	<code>heightmap_raster_export.m</code> - direct height-map STL/CSV to TXT exporter.
Contour STL exporter	<code>stl_slice_export_mm_woodpile_true_resample.m</code> - contour-slicing STL exporter for non-height-map cases.
General pipeline	<code>tppdlw_process.m</code> plus core/ helpers for STL/STEP imports and arbitrary scan angles.
Preview tools	<code>viz/preview_toolpath.m</code> , <code>viz/preview_3d.m</code> , and layer comparison tools.
Examples/tests	<code>examples/</code> and <code>tests/</code> show basic usage and expected output format.
Generated data	<code>output/</code> , STL files, and large TXT exports are ignored by Git by default.

2. Quick Start

From MATLAB, open the project folder and run:

```
cd('/path/to/3D_Printer_construct')
run('heightmap_raster_export.m')
```

- Set `HeightMapPath` to your input height-map STL or CSV file.
- Set `OutTxt` to the TXT file you want to create.
- Set `BaseHeight` if the source CSV needs a support base; leave it at 0 if the STL already includes one.
- Use `StageZConvention=false` for positive Z output, or the GUI `Z +` button next to `Offset Z / sign`.
- Start with coarse `XYPitch` and `DZ` for preview runs, then restore final values for production.
- Read the console summary. Height-map raster export should not report odd contour-crossing warnings.

3. Output Format

The output TXT is tab-separated, with one row per laser-on scan segment or Z transition:

```
X1    Y1    Z1    X2    Y2    Z2
```

- Rows where `Z1 == Z2` are scan/write segments.
- Rows where `Z1 != Z2` are layer-to-layer Z moves.
- Coordinates are in millimeters.
- The height-map raster exporter writes positive build height as negative stage Z when `StageZConvention` is true.

4. Main Parameters

Parameter	Purpose	Typical use
HeightMapPath	Input height-map STL or CSV file.	Use this path for pixel-exact height maps.
OutTxt	Destination tab-separated TXT file.	Use output/name.txt for generated files.
TargetMaxXY	Scales the model to a target XY footprint in mm.	Scalar for square fit, or [maxX maxY].
XYPitch	Spacing between adjacent raster scanlines.	Larger for preview, final value for production.
DZ	Layer thickness / vertical slicing step.	Larger for preview, final value for production.
PixelPitch	Source pixel pitch for CSV input.	STL can read this from the generated header.
BaseHeight	Support base height in source units.	Use 0 for STL that already includes a base; use 0.5 for a micron-valued CSV needing a 0.5 um base.
WoodpileMode	Alternates horizontal and vertical writing by layer.	Keep true for orthogonal resampling.
Serpentine	Alternates scan direction to reduce travel.	Usually true.
CoordMode	Uses voxel edges or grid centers for endpoints.	Use edges for full-width base rows.
StageZConvention	Controls output Z sign.	true writes negative stage Z; false writes positive Z, such as +0.00025.
OutputSignificant Digits	Controls compact TXT numeric precision.	Default 6 keeps values like 0.0002 readable.
Tolerance_mm	Height comparison tolerance in mm.	Default is usually fine.

5. Size and Pixel Pitch

- PixelPitch describes the original height-map pixel spacing in source units, such as 6 for a 6 um CSV pixel.
- TargetMaxXY controls the final maximum XY footprint in millimeters.
- XYPitch is the printer raster scanline spacing in millimeters. It controls write density and TXT size, not the source model footprint.
- If TargetMaxXY is set, the final source-pixel pitch is derived from target size and grid count. For a 335 x 335 map at 1.005 mm, one source pixel becomes $1.005 / 335 = 0.003$ mm.
- BaseHeight is added in heightmap_to_segments before raster scanlines are generated, so it creates full support-base scanlines across the height-map footprint.

6. Preview and Quality Checks

After generating a TXT file, preview a layer before printing:

```
addpath(genpath(pwd))
segments = load('output/Final.txt');
preview_toolpath(segments, 'Layer', 1, 'Mode', '2d')
preview_3d(segments, 'EveryN', 2)
```

Before sending a file to the printer, check for long strokes that cross empty space, out-of-bounds coordinates, unexpected zero-length rows, and unusually high segment counts on one layer.

If the first rows are short fragments along a boundary such as $X = 0$, a contour-slicing workflow is being used. Use `heightmap_raster_export.m` for pixel-exact height-map data so the file starts with hatch-fill rows.

Important: the contour-slicing fallback skips unresolved odd scanlines instead of force-pairing them. For pixel-exact height maps, prefer the raster workflow so contour crossings are not used at all.

7. Troubleshooting

Long lines across empty regions	Usually caused by using contour slicing on height-map data. Use height-map raster export first.
Very slow conversion	Increase <code>XPitch</code> and <code>DZ</code> for preview.
Missing small features	Check whether <code>XPitch</code> or <code>DZ</code> is too coarse.
TXT is huge	This is expected for fine pitch and large footprints. Keep generated TXT files under output/ so Git ignores them.
STL import fails	Confirm the STL is binary or valid ASCII, not truncated, and that the file path is correct.

8. Recommended Workflow

- **Preview pass:** use coarse pitch and layer height; generate quickly and inspect several layers.
- **Geometry check:** confirm the recovered height-map pitch, base height, and output span match the intended print.
- **Final pass:** restore production `XPitch` and `DZ`; regenerate TXT.
- **Visual QA:** preview first, middle, and last layers; check that no scanline crosses empty space.
- **Archive:** commit code/config changes, not large generated files. Keep STL/TXT outputs ignored unless you intentionally need to version them.

9. Quick Checklist Before Printing

[] Height-map source is correct	Right file, right units, expected base height and bounding box.
[] Parameters reviewed	Target size, XY pitch, DZ, coordinate mode, and output path are intentional.
[] Console warnings reviewed	No unexpected truncated STL, missing height-map cells, or out-of-bounds warnings.
[] Layers previewed	First, middle, and final layers look physically plausible.
[] Output format checked	TXT has six numeric columns and the expected number of layers.

[] Generated files managed

Large STL/TXT files are kept out of Git unless deliberately tracked.

For maintainers: keep the false-line prevention behavior conservative. Skipping an unresolved scanline is preferable to writing a long segment across empty space.